

Sales Performance Dashboard

Business Insights Report

Data Analytics Internship — Task 1

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Project Overview

This report analyzes transaction-level sales data from a UK-based online retailer (Online Retail II dataset) to uncover revenue trends, top-performing products, regional sales distribution, and customer purchasing behavior. The goal is to surface actionable business insights that could inform inventory, marketing, and regional expansion decisions.

Dataset & Methodology

The dataset covers invoice-level transactions including product descriptions, quantities, prices, customer IDs, and countries. Data cleaning removed cancelled orders and invalid entries (non-positive quantity or price), after which a Revenue field was derived as $\text{Quantity} \times \text{Price}$. Analysis was performed in Python (pandas, matplotlib) within a Jupyter notebook environment.

Core KPIs

Metric	Value
Total Revenue	£20,476,260.45
Total Orders	40,077
Total Unique Customers	5,878
Average Order Value	£510.92

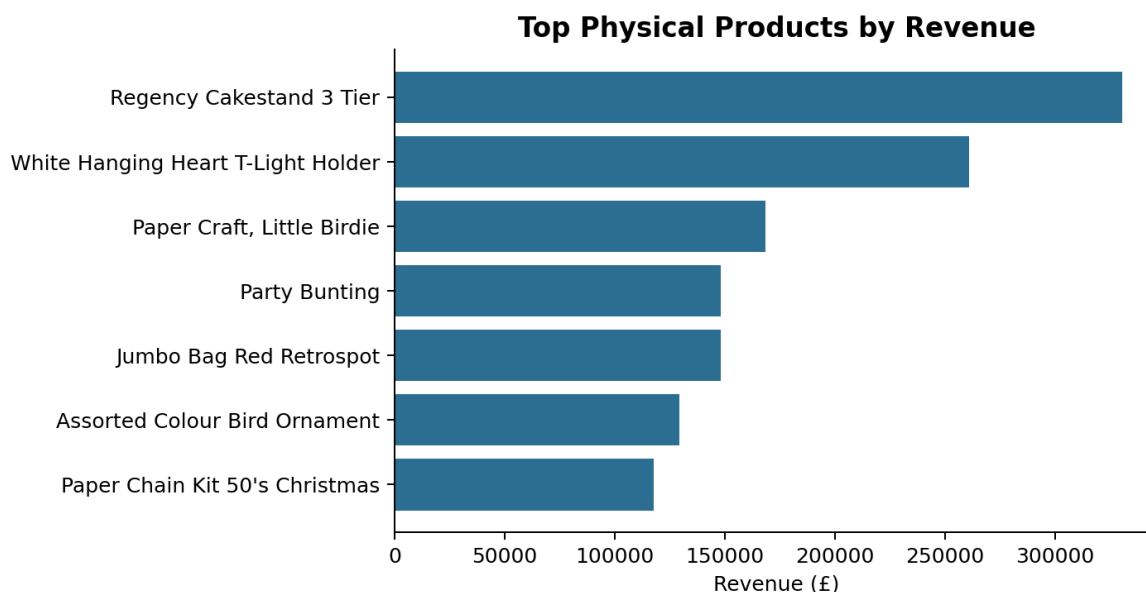
Revenue Trend

Monthly revenue shows a strong, recurring seasonal pattern. Revenue holds a relatively flat baseline of roughly £0.6M–£0.85M per month for most of the year, then spikes sharply each November — reaching approximately £1.45M in November 2010 — before falling back, with an even larger spike building toward November/December 2011. This pattern is consistent with pre-Christmas holiday buying behavior.

Insight: Inventory and staffing should be scaled up ahead of Q4, particularly October–November, to capture peak holiday demand. Off-peak months may benefit from targeted promotions to smooth revenue variability.

Top Products by Revenue

Among genuine physical products (excluding administrative line items like postage/manual adjustments), the Regency Cakestand 3 Tier (£330,590) and the White Hanging Heart T-Light Holder (£260,990) are the clear top performers, followed by a cluster of decorative and gift-oriented items.

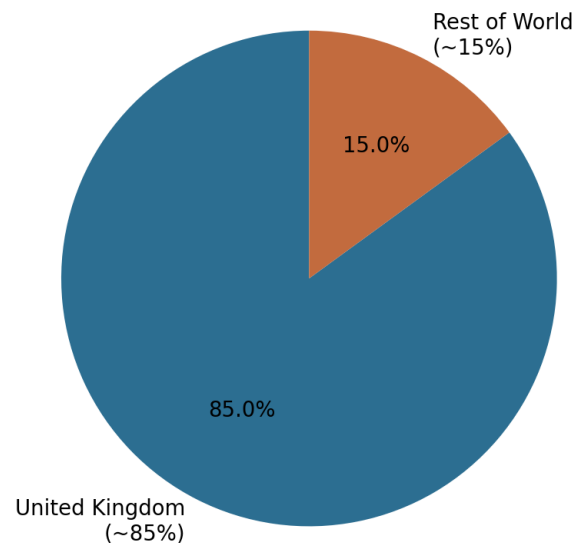


Insight: Decorative/gift items dominate top sales — inventory planning should prioritize stock depth for this category, especially ahead of the November–December peak.

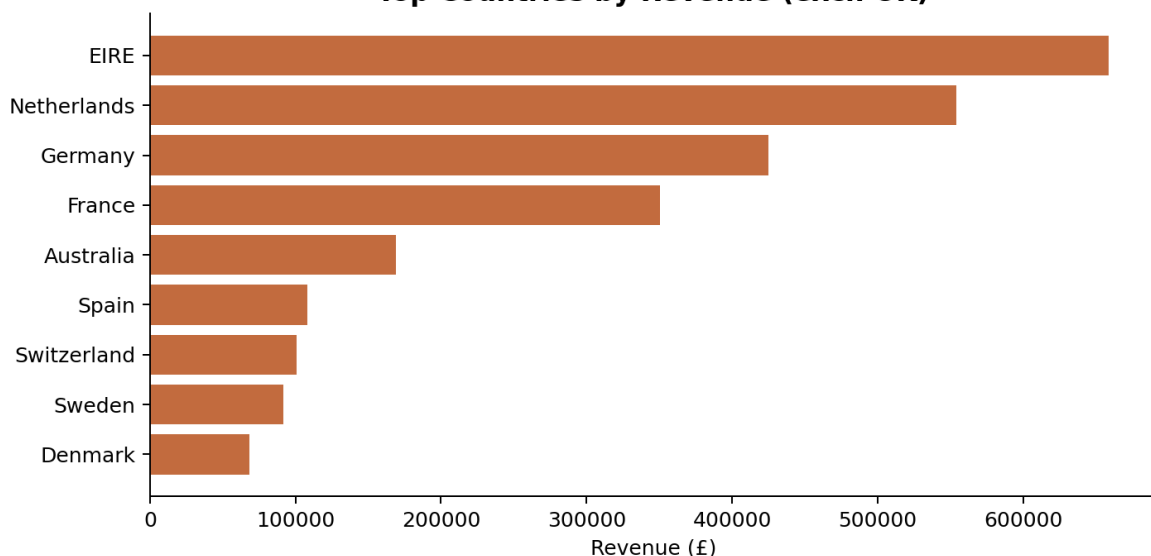
Regional Sales Performance

The United Kingdom accounts for approximately 85% of total revenue (£17.41M of £20.48M). EIRE (Ireland) is the largest international market at just 3.2% of revenue (£658,767), followed by the Netherlands, Germany, and France.

Revenue Share: UK vs Rest of World



Top Countries by Revenue (excl. UK)



Insight: Revenue is heavily concentrated in a single market, which represents concentration risk but also headroom — international markets are comparatively under-penetrated and could be a growth opportunity through targeted regional marketing.

Customer Purchasing Behavior

With 5,878 unique customers generating 40,077 orders, the average customer places roughly 6.8 orders — indicating that repeat purchasing plays a substantial role in overall revenue, rather than one-time transactions alone.

Insight: Given the meaningful role of repeat customers, targeted retention strategies (loyalty incentives, personalized offers) are likely to be high-leverage. This finding directly motivates the customer segmentation analysis carried out in Task 2.

Summary of Key Insights

- Revenue is strongly seasonal, peaking each November ahead of the holiday season.
- Decorative and gift-oriented products (e.g. Regency Cakestand 3 Tier, White Hanging Heart T-Light Holder) are the strongest physical-product performers.
- The UK contributes ~85% of total revenue, with EIRE, the Netherlands, and Germany as the leading (but comparatively small) international markets.
- Customers order an average of ~6.8 times each, suggesting repeat purchasing is a core revenue driver worth targeting through retention strategies.

Recommendations

- Scale inventory and marketing spend ahead of Q4 to capture peak seasonal demand.
- Prioritize stock depth for top decorative/gift categories heading into the holiday period.
- Invest in targeted international marketing to grow underdeveloped markets outside the UK.
- Develop retention programs (loyalty offers, personalized email campaigns) aimed at repeat customers, informed further by the Task 2 segmentation analysis.